SOCIOL 723: Social Statistics II

Week 1, The Linear Model and OLS

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Today

Orientation

Three Pillars of this Course

- **Estimation** (Weeks 1–4). What is the estimator, and what are its statistical properties? OLS, then maximum likelihood as the general engine.
- **Causal inference** (Weeks 5–11). What quantity are we after, and what would have to be true for our estimate to recover it?
- **Machine learning** (Weeks 12–13). What do we do when the number of covariates is large and the functional form is unknown?

The through-line

Nearly everything in this course is some version of one idea: *compare units that are alike in all the ways that matter*. Regression is the first and most transparent implementation of that idea.

How to Read These Slides

The convention

A frame title marked with ★ contains material that is here *for your understanding*, not for evaluation. Derivations in matrix form, projection geometry, and asymptotic arguments all fall in this category.

- **What I will examine**: the scalar versions of everything, the algebra you can do by hand with $\sum_i$ signs, plus interpretation, assumptions, and knowing which tool fits which problem.
- **What is here anyway**: the general, abstract versions. You should read them, and you will meet them again in the literature. You will not be asked to reproduce them under time pressure.
- My slides are deliberately dense so that they work as a reference after class. Stop me whenever something is unclear; that is what lecture is for.

What You Already Know: Model Comparison

In *Social Statistics I* you compared a **compact** model to an **augmented** model:

$$\begin{aligned} \text{MODEL C:} \quad & Y_i = \beta_0 + \beta_1 X_{1i} + \epsilon_i \\ \text{MODEL A:} \quad & Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i \end{aligned}$$

and asked whether the extra parameters bought enough reduction in error:

$$PRE = \frac{SSE_C - SSE_A}{SSE_C}, \qquad F = \frac{(SSE_C - SSE_A)/(PA - PC)}{SSE_A/(n - PA)}.$$

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- All of this stays. Today we look *inside* the estimator: where do $\hat{\beta}_0, \hat{\beta}_1$ come from, and what do they mean?
- Then we rewrite it in matrix form, which is the language the rest of the course speaks.

Notation We Will Use All Semester

Population versus sample

$$E[\,\cdot\,] : \text{the } \textbf{population} \text{ expectation, a fixed, unknown number}$$

$$\mathbb{E}_n[\,\cdot\,] \equiv \frac{1}{n}\sum_{i=1}^{n}(\,\cdot\,) : \text{the } \textbf{sample} \text{ (empirical) expectation, a random variable}$$

- Greek letters (β, σ, ϵ) are *population* quantities we never observe; hats ($\hat{\beta}$, $\hat{e}$) are *sample* quantities we compute.
- ϵ_i always denotes the error term; $\hat{e}_i$ the fitted residual. They are different objects, and confusing them is the single most common source of trouble.
- Later: bold italic = vector or matrix (lowercase $\boldsymbol{y}$ for vectors, uppercase $\boldsymbol{X}$ for matrices), $'$ = transpose.

The whole of statistical inference is the study of the gap between E and $\mathbb{E}_n$.

Scalar OLS

The Bivariate Model

The Bivariate Linear Model

For each unit $i = 1, \ldots, n$ we observe an outcome Y_i and one regressor X_i:

$$Y_i = \beta_0 + \beta_1 X_i + \epsilon_i.$$

- β_0, the **intercept**: the average of Y when $X = 0$.
- β_1, the **slope**: how much higher Y is, on average, for units that differ by one unit in X.
- ϵ_i, the **error**: everything else that makes unit i's outcome differ from the line.

Running example, all semester

GSS 2010–2022, full-time workers aged 25–64 ($n = 3{,}509$).
Y_i = log personal earnings, X_i = years of schooling completed.

The Least Squares Criterion

We do not observe β_0, β_1. We pick estimates b_0, b_1 to make the fitted line as close as possible to the data. "Close" = small squared vertical distances:

$$S(b_0, b_1) \;=\; \sum_{i=1}^{n} \big(Y_i - b_0 - b_1 X_i\big)^2.$$

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- Why squares? They punish large misses more than small ones, they make the problem differentiable, and, as we will see, they deliver the conditional *mean*.
- Why vertical? Because we are predicting Y from X, not the reverse. Regressing X on Y gives a different line.

The OLS estimates are $(\hat{\beta}_0, \hat{\beta}_1) = \arg\min_{b_0, b_1} S(b_0, b_1)$.

Step 1: The First-Order Conditions

S is a smooth, convex function of two numbers. Set both partial derivatives to zero. Using the chain rule,

$$\frac{\partial S}{\partial b_0} = \sum_{i=1}^{n} 2(Y_i - b_0 - b_1 X_i) \cdot (-1) \overset{!}{=} 0,$$

$$\frac{\partial S}{\partial b_1} = \sum_{i=1}^{n} 2(Y_i - b_0 - b_1 X_i) \cdot (-X_i) \overset{!}{=} 0.$$

Dividing by -2 gives the two normal equations:

$$\sum_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0, \tag{N1}$$

$$\sum_i X_i (Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i) = 0. \tag{N2}$$

What the Normal Equations Say

Write $\hat{e}_i = Y_i - \hat{\beta}_0 - \hat{\beta}_1 X_i$ for the fitted residual. Then

$$\text{(N1)}: \quad \sum_i \hat{e}_i = 0 \qquad\qquad \text{(N2)}: \quad \sum_i X_i \hat{e}_i = 0.$$

- The residuals **sum to zero**. Hence $\bar{\hat{e}} = 0$, and the fitted line passes through the point of means $(\bar{X}, \bar{Y})$.
- The residuals are **uncorrelated with the regressor**: given (N1), (N2) is equivalent to $\sum_i (X_i - \bar{X})\hat{e}_i = 0$.

Read this carefully

These are *mechanical consequences of the algebra*, true in every sample, for every dataset. Plotting $\hat{e}_i$ against X_i and finding no correlation is therefore not evidence that the model is correct.

Step 2: Solving for $\hat{\beta}_0$

Divide (N1) by n:

$$\frac{1}{n}\sum_i Y_i - \hat{\beta}_0 - \hat{\beta}_1 \frac{1}{n}\sum_i X_i = 0 \quad \Longrightarrow \quad \bar{Y} = \hat{\beta}_0 + \hat{\beta}_1 \bar{X},$$

so

$$\boxed{\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}}$$

- The intercept is whatever it must be for the line to pass through $(\bar{X}, \bar{Y})$.
- Immediate consequence: if you *center* the regressor ($X_i^c = X_i - \bar{X}$), the intercept becomes $\bar{Y}$. Centering never changes the slope, only what the intercept means.

Step 3: Solving for $\hat{\beta}_1$

Substitute $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$ into (N2):

$$\sum_i X_i \Big(Y_i - \bar{Y} + \hat{\beta}_1 \bar{X} - \hat{\beta}_1 X_i \Big) = 0$$

$$\sum_i X_i (Y_i - \bar{Y}) = \hat{\beta}_1 \sum_i X_i (X_i - \bar{X}).$$

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Now use $\sum_i \bar{X}(Y_i - \bar{Y}) = 0$ and $\sum_i \bar{X}(X_i - \bar{X}) = 0$, which let us replace X_i by $(X_i - \bar{X})$ on both sides:

$$\boxed{\hat{\beta}_1 = \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{\mathrm{Cov}_n(X, Y)}{\mathrm{Var}_n(X)}}$$

Reading the Slope Formula

$$\hat{\beta}_1 = \frac{\text{Cov}_n(X,Y)}{\text{Var}_n(X)} = \widehat{\text{Corr}}(X,Y) \cdot \frac{s_Y}{s_X} = \sum_i w_i \, \frac{Y_i - \bar{Y}}{X_i - \bar{X}}, \quad w_i = \frac{(X_i - \bar{X})^2}{\sum_j (X_j - \bar{X})^2}.$$

- **Numerator**: do X and Y move together across units?
- **Denominator**: how much does X vary? No variation in X, no slope; the formula is $0/0$.
- The third form is worth remembering: the OLS slope is a *weighted average of unit-by-unit slopes*, with more weight on observations whose X is far from $\bar{X}$. Extreme X values matter disproportionately.

The Running Example, by Hand

GSS 2010–2022, $n = 3{,}509$, $Y =$ log earnings, $X =$ years of schooling:

$$\bar{X} = 14.48, \quad \bar{Y} = 9.950, \quad \mathrm{Cov}_n(X, Y) = 0.9476, \quad \mathrm{Var}_n(X) = 7.981.$$

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$$\begin{aligned}\hat{\beta}_1 &= \frac{0.9476}{7.981} = 0.1187, \\ \hat{\beta}_0 &= 9.950 - 0.1187 \times 14.48 = 8.231.\end{aligned}$$

Interpretation

Comparing two full-time workers who differ by one year of schooling, the one with more schooling has about 11.9% higher earnings *on average*.

Note the words: *comparing*, *on average*. Not "one more year **causes** 11.9% higher earnings."

Interpretation: Units, Logs, and Nonlinearity

- Y in levels, X in levels: β_1 is a change in Y-units per X-unit.
- $\log Y$, X **in levels** (our case): $\beta_1 \approx$ the *proportional* change in Y. Exactly, $100(e^{\beta_1} - 1)\%$; for $\beta_1 = 0.1187$ this is 12.6%. The approximation is good below about 0.2.
- $\log Y$, $\log X$: β_1 is an elasticity.
- **Quadratics**: with $Y_i = \beta_0 + \beta_1 X_i + \beta_2 X_i^2 + \epsilon_i$, the marginal effect is $\beta_1 + 2\beta_2 X$, it depends on where you evaluate it. Report it at meaningful values, not only at the mean.
- **Interactions**: with $\beta_3 X_i Z_i$, the effect of X is $\beta_1 + \beta_3 Z$. The "main effect" β_1 is the effect at $Z = 0$, which may be outside the data.

Multiple Regression, in Scalar Form

Adding a Second Regressor

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \epsilon_i.$$

Minimizing $S(b_0, b_1, b_2) = \sum_i (Y_i - b_0 - b_1 X_{1i} - b_2 X_{2i})^2$ gives *three* normal equations:

$$\sum_i \hat{e}_i = 0, \qquad \sum_i X_{1i}\hat{e}_i = 0, \qquad \sum_i X_{2i}\hat{e}_i = 0.$$

- One equation per parameter. With k parameters you get k equations in k unknowns, which is exactly why the matrix form will be so much more compact.
- Solving three equations by substitution is unpleasant but possible; with ten it is hopeless. That is the whole motivation for matrices.

What "Holding Constant" Actually Means

$\hat{\beta}_1$ is usually described as "the effect of X_1 *holding X_2 constant.*" Where does that come from?

Partialling out: a three-step recipe

1. Regress X_{1i} on X_{2i} (and a constant). Keep the residuals $\tilde{X}_{1i} = X_{1i} - \hat{\gamma}_0 - \hat{\gamma}_1 X_{2i}$.
2. Regress Y_i on X_{2i}. Keep the residuals $\tilde{Y}_i$.
3. Regress $\tilde{Y}_i$ on $\tilde{X}_{1i}$. The slope you get is exactly $\hat{\beta}_1$ from the multiple regression.

This is the **Frisch–Waugh–Lovell** theorem. In scalar form:

$$\hat{\beta}_1 = \frac{\sum_i \tilde{X}_{1i}\,\tilde{Y}_i}{\sum_i \tilde{X}_{1i}^2} = \frac{\sum_i \tilde{X}_{1i}\,Y_i}{\sum_i \tilde{X}_{1i}^2}.$$

Why This Matters (cont.)

- "Controlling for X_2" means throwing away every scrap of X_1 that X_2 can predict, and using only what is left.
- If X_1 and X_2 are strongly related, very little is left. The coefficient is then estimated from a thin, possibly unrepresentative slice of the data, and its standard error will say so.
- In our data $sd(\texttt{educ}) = 2.82$; after residualizing on experience it is 2.73. Little is lost, so the two coefficients on `educ` are similar (0.119 vs. 0.138).
- You may residualize Y or not; the slope is the same either way, because the part removed from Y is a function of X_2 and is therefore orthogonal to $\tilde{X}_1$.
- What you cannot change is the denominator: it must be $\sum_i \tilde{X}_{1i}^2$, not $\sum_i (X_{1i} - \bar{X}_1)^2$. Regressing $\tilde{Y}$ on *raw* X_1 divides by too large a number and attenuates the estimate. (Problem Set 1.)

Why This Matters (cont.)

The plot of $\tilde{Y}$ against $\tilde{X}_1$, the *added-variable plot*, is the only honest two-dimensional picture of a multivariate slope.

Omitted Variables

Short and Long Regressions

Suppose the "long" regression is

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_i + \epsilon_i,$$

but we run the "short" one, omitting W_i:

$$Y_i = \tilde{\beta}_0 + \tilde{\beta}_1 X_i + u_i.$$

Let δ_1 be the slope from the *auxiliary* regression of the omitted variable on the included one: $W_i = \delta_0 + \delta_1 X_i + \nu_i$.

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Omitted variable bias

$$\boxed{\tilde{\beta}_1 \ = \ \beta_1 \ + \ \beta_2 \, \delta_1}$$

"short = long + (effect of omitted) × (regression of omitted on included)"

Derivation, in Scalar Form

Apply the bivariate slope formula to the short regression, then substitute the long model for Y_i. Everything below is *sample* projection algebra: the long regression's residual is orthogonal to its own regressors by construction, which is why the last term is exactly zero.

$$
\begin{aligned}
\tilde{\beta}_1 &= \frac{\mathrm{Cov}_n(X,Y)}{\mathrm{Var}_n(X)} = \frac{\mathrm{Cov}_n\left(X,\ \beta_0+\beta_1 X+\beta_2 W+\epsilon\right)}{\mathrm{Var}_n(X)} \\
&= \beta_1 \underbrace{\frac{\mathrm{Cov}_n(X,X)}{\mathrm{Var}_n(X)}}_{=1} + \beta_2 \underbrace{\frac{\mathrm{Cov}_n(X,W)}{\mathrm{Var}_n(X)}}_{=\delta_1} + \underbrace{\frac{\mathrm{Cov}_n(X,\epsilon)}{\mathrm{Var}_n(X)}}_{=0 \text{ in the long regression}} \\
&= \beta_1 + \beta_2\,\delta_1. \qquad \blacksquare
\end{aligned}
$$

Constants drop out of covariances, which is why β_0 never appears.

Signing the Bias

	$\mathrm{Cov}_n(X, W) > 0$	$\mathrm{Cov}_n(X, W) < 0$
$\beta_2 > 0$	positive bias (*overstate*)	negative bias (*understate*)
$\beta_2 < 0$	negative bias (*understate*)	positive bias (*overstate*)

- You need *both* channels. An omitted variable uncorrelated with X costs precision, not consistency.
- Omit ability from a schooling–earnings regression: ability raises earnings ($\beta_2 > 0$) and rises with schooling ($\delta_1 > 0$) $\Rightarrow$ the return to schooling is overstated.
- With several omitted variables you cannot generally sign the total bias from theory alone; the terms can offset.

Omitted Variable Bias in the GSS

Log earnings on schooling; full-time workers 25–64, $n = 3{,}509$.

	(1)	(2)	(3)
Years of schooling	0.119	0.140	0.118
Experience, experience^2, female, Black		✓	✓
Vocabulary score (WORDSUM), parental education			✓
R^2	0.129	0.233	0.245

Check the formula on column (3) → (2). The two omitted variables contribute

$$\underbrace{0.0406}_{\hat{\beta}_{\text{wordsum}}} \times \underbrace{0.281}_{\hat{\delta}_{\text{wordsum}}} + \underbrace{0.0205}_{\hat{\beta}_{\text{pareduc}}} \times \underbrace{0.527}_{\hat{\delta}_{\text{pareduc}}} = 0.0222 = 0.1405 - 0.1183. \quad \checkmark$$

But how much is *still* missing? The algebra is silent. That question is what Weeks 5–11 are about.

Fit

Decomposing the Variation

Write $\hat{Y}_i = \hat{\beta}_0 + \hat{\beta}_1 X_i$ so that $Y_i = \hat{Y}_i + \hat{e}_i$. Subtract $\bar{Y}$, square, and sum:

$$\underbrace{\sum_i (Y_i - \bar{Y})^2}_{TSS = SSE_C} = \underbrace{\sum_i (\hat{Y}_i - \bar{Y})^2}_{ESS} + \underbrace{\sum_i \hat{e}_i^2}_{RSS = SSE_A} + 2 \underbrace{\sum_i (\hat{Y}_i - \bar{Y}) \hat{e}_i}_{= 0}.$$

The cross term vanishes because $\sum_i \hat{e}_i = 0$ and $\sum_i X_i \hat{e}_i = 0$, and $\hat{Y}_i$ is a linear function of X_i.

$$R^2 = \frac{ESS}{TSS} = 1 - \frac{RSS}{TSS} \qquad \Longleftrightarrow \qquad PRE = \frac{SSE_C - SSE_A}{SSE_C}.$$

R^2 is exactly the PRE of your model against the intercept-only model. In our example $TSS = 3062.3$, $RSS = 2667.6$, so $R^2 = 0.129$.

A Warning About R^2

- R^2 never decreases when you add a regressor. Adjusted R^2 penalizes k, but is still not a principled model-selection rule.
- R^2 depends on how much X varies *in your sample*, which is a feature of the sampling design, not of the relationship.
- For causal work R^2 is nearly irrelevant: a randomized experiment with $R^2 = 0.01$ identifies the effect; an observational regression with $R^2 = 0.9$ identifies nothing in particular.

Use it for

Prediction quality, and comparing nested specifications on the *same* sample. Not for adjudicating causal claims.

Checkpoint: What You Should Be Able to Do

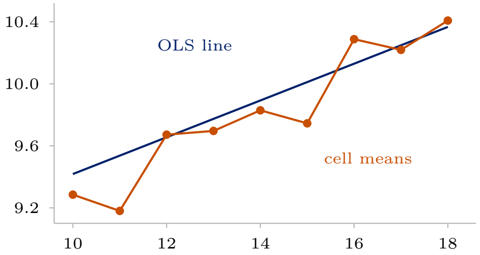
1. Write down the least-squares objective and differentiate it to get the normal equations.
2. State what the normal equations imply about the residuals, and explain why those implications are not evidence of anything.
3. Derive and interpret $\hat{\beta}_1 = \mathrm{Cov}_n(X, Y) / \mathrm{Var}_n(X)$ and $\hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}$.
4. Describe partialling out in three steps and say what "holding constant" costs.
5. Derive $\tilde{\beta}_1 = \beta_1 + \beta_2 \delta_1$ and sign the bias.
6. Connect R^2 to PRE.

Next we ask the question all of this has been dodging: what is $\hat{\beta}_1$ an estimate *of*?

CEF and BLP

Nobody Believes the Line

Regress `lnearn` on `educ` in our GSS extract: $\hat{\beta}_1 = 0.119$. Now plot the *actual* mean of `lnearn` at each value of `educ`.



step	actual	the line says
$11 \to 12$	$+0.49$	$+0.12$
$12 \to 13$	$+0.02$	$+0.12$
$15 \to 16$	$+0.54$	$+0.12$
$16 \to 17$	-0.07	$+0.12$

The returns are concentrated at the diploma years, 12 and 16. The line spreads them evenly across all years. It misses the shape entirely: it has no idea that 12 and 16 are special.

And yet we reported 0.119 anyway. What is that number?

Estimating *What*, Exactly?

Look carefully at how we defined this estimator's target.

The circularity

We *wrote down* $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$ and then showed $E[\hat{\beta}_1] = \beta_1$. But β_1 exists only because we assumed that equation. If the CEF is not a line, and it is not, there is no true β_1 for $\hat{\beta}_1$ to be unbiased *for*.

- **Estimand**: the population quantity you want. **Estimator**: the recipe. **Estimate**: the number.
- We have the last two. We have never named the first without assuming it into existence.
- Unbiasedness, consistency, standard errors are all statements about distance from a target. No target, no statements.

The Plan

Define β as a feature of the joint distribution of (Y_i, X_i) , something that exists in every population, whether or not any line is true.

1. The **CEF** $E[Y_i \mid X_i]$: what we would most like to know. Assumption-free, and the answer to most sociological questions as actually posed.
2. The **best linear predictor**: the line closest to the CEF. Defined for any joint distribution, no linearity assumed, none needed.
3. A theorem connecting them, which tells us exactly what 0.119 is a summary *of*.

Why this is worth an hour

It converts "your model is misspecified" from a fatal objection into a statement about *approximation error*, which you can then evaluate, bound, or fix by saturating. You cannot defend a regression you cannot say the target of.

The Conditional Expectation Function (CEF)

Everything so far was about a *sample*. Step back to the population.

For an outcome Y_i and covariates X_i, the conditional expectation function is

$$g(x) \;=\; E[Y_i \mid X_i = x].$$

- In words: the average outcome among all units in the population with $X_i = x$, average earnings among people with exactly 16 years of schooling.
- This is what most sociological questions are really about: *how does the average of Y move as we move through the population along X?*
- Because X_i is random, $E[Y_i \mid X_i]$ is itself a random variable.

Three Facts About the CEF

1. Law of iterated expectations

$E[Y_i] = E\big[E[Y_i \mid X_i]\big]$, the overall mean is the average of the group means.

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2. CEF decomposition

Define $\epsilon_i \equiv Y_i - E[Y_i \mid X_i]$. Then $Y_i = E[Y_i \mid X_i] + \epsilon_i$ with $E[\epsilon_i \mid X_i] = 0$, and hence $E[\epsilon_i m(X_i)] = 0$ for *any* function $m(\cdot)$.

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3. CEF prediction property

$E[Y_i \mid X_i] = \arg\min_{m(\cdot)} E\big[(Y_i - m(X_i))^2\big]$: among *all* functions of X_i, the CEF is the best mean-squared-error predictor.

These are identities, not assumptions. They carry no causal content whatever.

Proof of the Decomposition Property ★

By definition $\epsilon_i = Y_i - E[Y_i \mid X_i]$, so

$$E[\epsilon_i \mid X_i] = E\big[Y_i \mid X_i\big] - E[Y_i \mid X_i] = 0.$$

For any function $m(\cdot)$, apply the law of iterated expectations:

$$E\big[\epsilon_i\, m(X_i)\big] = E\Big[E\big[\epsilon_i\, m(X_i) \mid X_i\big]\Big] = E\Big[m(X_i)\, \underbrace{E[\epsilon_i \mid X_i]}_{=0}\Big] = 0. \quad \blacksquare$$

Any random variable splits into a part *explained* by X_i and a part orthogonal to *every* function of X_i.

Proof of the Prediction Property ★

For any candidate predictor $m(X_i)$, add and subtract the CEF:

$$
\begin{aligned}
E\left[(Y_i - m(X_i))^2\right] &= E\left[\left\{(Y_i - E[Y_i \mid X_i]) + (E[Y_i \mid X_i] - m(X_i))\right\}^2\right] \\
&= \underbrace{E\left[(Y_i - E[Y_i \mid X_i])^2\right]}_{\text{does not involve } m} + \underbrace{E\left[(E[Y_i \mid X_i] - m(X_i))^2\right]}_{\geq 0} \\
&\quad + 2\, \underbrace{E\left[\epsilon_i \left(E[Y_i \mid X_i] - m(X_i)\right)\right]}_{=0 \text{ by decomposition}}.
\end{aligned}
$$

The middle term is minimized at zero by setting $m(X_i) = E[Y_i \mid X_i]$. ■

The Population Regression, or Best Linear Predictor

The CEF is a general function and typically unknown. Restrict attention to *linear* predictors and define

$$\boldsymbol{\beta} = \arg\min_{b} E\left[(Y_i - X_i' b)^2\right] \quad \Longrightarrow \quad E\left[X_i(Y_i - X_i'\boldsymbol{\beta})\right] = 0.$$

This is the population counterpart of the normal equations. In the bivariate case it is exactly what you expect:

$$\beta_1 = \frac{\mathrm{Cov}(X_i, Y_i)}{\mathrm{Var}(X_i)}, \qquad \beta_0 = E[Y_i] - \beta_1 E[X_i].$$

Defining $e_i \equiv Y_i - X_i'\boldsymbol{\beta}$, we get $E[X_i e_i] = 0$ by construction, in every population. Again: algebra, not causality.

Why Should We Care About the Best *Linear* Predictor?

Three standard justifications (Angrist and Pischke 2009):

1. **If the CEF is linear**, the population regression *is* the CEF.
2. **If the CEF is nonlinear**, $X_i'\boldsymbol{\beta}$ is the minimum-MSE linear approximation to it: $\boldsymbol{\beta} = \arg\min_b E\big[(E[Y_i \mid X_i] - X_i'b)^2\big]$.
3. **Regardless**, $X_i'\boldsymbol{\beta}$ is the minimum-MSE linear predictor of Y_i.

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3. **Regardless**, $X_i'\boldsymbol{\beta}$ is the minimum-MSE linear predictor of Y_i.

Justification 2 is the useful one in practice. OLS is a *well-defined summary of the CEF* even when you do not believe the CEF is a line, so "the model is wrong" is not, by itself, a fatal objection.

Proof of the Regression-CEF Theorem ★

Claim: $\boldsymbol{\beta} = \arg\min_b E[(Y_i - X_i'b)^2]$ also solves $\arg\min_b E\left[(E[Y_i \mid X_i] - X_i'b)^2\right]$.

$$
\begin{aligned}
E[(Y_i - X_i'b)^2] &= E\Big[\big\{(Y_i - E[Y_i \mid X_i]) + (E[Y_i \mid X_i] - X_i'b)\big\}^2\Big] \\
&= \underbrace{E\left[(Y_i - E[Y_i \mid X_i])^2\right]}_{\text{free of } b} + E\left[(E[Y_i \mid X_i] - X_i'b)^2\right] \\
&\quad + 2\underbrace{E\left[\epsilon_i\,(E[Y_i \mid X_i] - X_i'b)\right]}_{=0}
\end{aligned}
$$

The two objectives differ by a constant, so they share a minimizer. ■

Saturated Models: When Regression Is the CEF

Suppose X_i is *discrete* with J cells. Include a full set of $J-1$ dummies plus a constant. The regression is then saturated: one free parameter per cell.

- A saturated regression reproduces the CEF *exactly*, cell by cell. No functional-form assumption is being made at all.
- Example: regress log earnings on a dummy for each value of `educ`. The fitted values *are* the cell means.
- Contrast: entering `educ` linearly imposes a constant increment, and is therefore an approximation.

Practical upshot

Whenever you can afford the degrees of freedom, saturating in the key covariate is a free robustness check on functional form. We do this in lab.

What That Detour Bought Us: Two Questions, Kept Apart

Every argument about a regression coefficient is one of two questions. Keep them apart and most confusion dissolves.

Q1. What does $\hat{\beta}_1$ estimate?

The minimum-MSE linear approximation to the CEF.

Always answerable, algebra, true in every population.

"Your model is misspecified" attacks this, and usually *fails*: OLS has an honest answer.

Q2. Is *that* the causal effect?

Only if the units being compared are comparable.

Never answerable by algebra, a claim about how the data came to be.

"Your treated and controls differ" attacks this, and usually *lands*: nothing can repair it.

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Weeks 1–4 are Q1. Weeks 5–13 are Q2. That is the whole course.

Matrix Form

Why Bother

- With k regressors, scalar algebra needs k normal equations and becomes unreadable. Matrix algebra stays two lines long for any k.
- The geometry, projection onto a subspace, is invisible in scalar form and obvious in matrix form.
- Every estimator later in the course (fixed effects, IV, 2SLS, ridge, weighting, double ML) is a small variation on $\hat{\beta} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y}$. You will not be able to read the literature without it.

But

Nothing in this section is new content. It is the *same* normal equations you already derived, written more compactly. Thursday's lab reviews every piece of linear algebra we use.

Stacking the Data

For n units and k regressors (including the constant):

$$\boldsymbol{y} = \begin{bmatrix} Y_1 \\ \vdots \\ Y_n \end{bmatrix}_{n\times 1}, \quad \boldsymbol{X} = \begin{bmatrix} X_1' \\ \vdots \\ X_n' \end{bmatrix} = \begin{bmatrix} 1 & X_{11} & \cdots & X_{1,k-1} \\ \vdots & \vdots & & \vdots \\ 1 & X_{n1} & \cdots & X_{n,k-1} \end{bmatrix}_{n\times k}, \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \vdots \\ \epsilon_n \end{bmatrix}.$$

The model, stacked over units:

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \qquad \Longleftrightarrow \qquad Y_i = X_i'\boldsymbol{\beta} + \epsilon_i, \quad i = 1, \ldots, n.$$

Two objects do all the work: $\boldsymbol{X}'\boldsymbol{X} = \sum_i X_i X_i'$ (a $k \times k$ matrix) and $\boldsymbol{X}'\boldsymbol{y} = \sum_i X_i Y_i$ (a $k \times 1$ vector).

Least Squares in Matrix Form

$$S(b) = \sum_i (Y_i - X_i' b)^2 = (\boldsymbol{y} - \boldsymbol{X} b)'(\boldsymbol{y} - \boldsymbol{X} b) = \boldsymbol{y}'\boldsymbol{y} - 2\, b'\boldsymbol{X}'\boldsymbol{y} + b'\boldsymbol{X}'\boldsymbol{X} b.$$

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Differentiate, using $\partial(a'b)/\partial b = a$ and $\partial(b'Ab)/\partial b = 2Ab$ for symmetric A:

$$\frac{\partial S(b)}{\partial b} = -2\boldsymbol{X}'\boldsymbol{y} + 2\boldsymbol{X}'\boldsymbol{X}b \stackrel{!}{=} 0 \quad \Longrightarrow \quad \underbrace{\boldsymbol{X}'\boldsymbol{X}\,\hat{\beta} = \boldsymbol{X}'\boldsymbol{y}}_{\text{normal equations}}.$$

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These are the *same* k equations you wrote out one at a time. The second-order condition holds because $\boldsymbol{X}'\boldsymbol{X} \succeq 0$, so S is convex and any stationary point is a global minimum.

The Estimator, Existence, and Rank

The OLS estimator

If $\operatorname{rank}(\boldsymbol{X}) = k$ (*full column rank*), $\boldsymbol{X}'\boldsymbol{X}$ is invertible and

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} = \left(\mathbb{E}_n[X_i X_i']\right)^{-1}\mathbb{E}_n[X_i Y_i].$$

- Full column rank requires $n \geq k$ and no exact linear dependence among columns. It is the matrix version of "X must vary."
- Failures you will actually meet: the *dummy variable trap*; including a variable and a rescaling of it; a group indicator constant within a fixed effect.
- If rank fails, `R` silently drops a column and reports `NA`. That `NA` is never a mystery.

Sanity Check: Recovering the Scalar Formula ★

Let $k = 2$ with $X_i = (1, X_i)'$. Then

$$\boldsymbol{X}'\boldsymbol{X} = \begin{bmatrix} n & \sum_i X_i \\ \sum_i X_i & \sum_i X_i^2 \end{bmatrix}, \qquad \boldsymbol{X}'\boldsymbol{y} = \begin{bmatrix} \sum_i Y_i \\ \sum_i X_i Y_i \end{bmatrix}.$$

Using $\begin{bmatrix} a & b \\ c & d \end{bmatrix}^{-1} = \frac{1}{ad-bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$,

$$\hat{\beta}_1 = \frac{n \sum_i X_i Y_i - \sum_i X_i \sum_i Y_i}{n \sum_i X_i^2 - (\sum_i X_i)^2} = \frac{\mathrm{Cov}_n(X, Y)}{\mathrm{Var}_n(X)}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1 \bar{X}.$$

Exactly what we derived by hand. The matrix form is a change of language, not of substance.

OLS Is a Projection ★

Think of $\boldsymbol{y}$ as a single point in $\mathbb{R}^n$, one dimension per *observation*. The k columns of $\boldsymbol{X}$ span a k-dimensional subspace

$$\text{col}(\boldsymbol{X}) = \{\boldsymbol{X}b : b \in \mathbb{R}^k\} \subset \mathbb{R}^n.$$

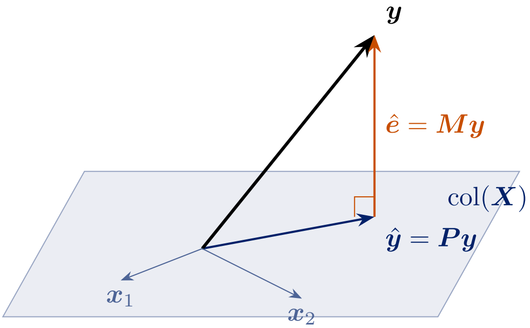
OLS finds the point in $\text{col}(\boldsymbol{X})$ closest to $\boldsymbol{y}$: $\hat{\boldsymbol{y}} = \arg\min_{v \in \text{col}(\boldsymbol{X})} \|\boldsymbol{y} - v\|^2$.

Projection and annihilator matrices

$$\boldsymbol{P} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' \text{ (``hat matrix''),} \qquad \boldsymbol{M} = \boldsymbol{I}_n - \boldsymbol{P} \text{ (``residual maker'')}$$

so that $\hat{\boldsymbol{y}} = \boldsymbol{P}\boldsymbol{y}$ and $\hat{\boldsymbol{e}} = \boldsymbol{M}\boldsymbol{y}$.

The Picture ★



- $\hat{\boldsymbol{e}} \perp \text{col}(\boldsymbol{X})$, the geometric content of $\boldsymbol{X}'\hat{\boldsymbol{e}} = \mathbf{0}$, which is the geometric content of "residuals are uncorrelated with the regressors."
- Fitting is dropping a perpendicular. Everything else is bookkeeping.

Properties of P and M ★

Symmetric

$$\boldsymbol{P}' = \boldsymbol{P}, \qquad \boldsymbol{M}' = \boldsymbol{M}$$

Idempotent

$$\boldsymbol{P}\boldsymbol{P} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\underbrace{\boldsymbol{X}'\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}}_{\boldsymbol{I}}\boldsymbol{X}' = \boldsymbol{P}$$

$$\boldsymbol{M}\boldsymbol{M} = (\boldsymbol{I} - \boldsymbol{P})(\boldsymbol{I} - \boldsymbol{P}) = \boldsymbol{M}$$

Mutually orthogonal

$$\boldsymbol{P}\boldsymbol{M} = \boldsymbol{P} - \boldsymbol{P}\boldsymbol{P} = \boldsymbol{0}$$

Act correctly on $\boldsymbol{X}$

$$\boldsymbol{P}\boldsymbol{X} = \boldsymbol{X}, \qquad \boldsymbol{M}\boldsymbol{X} = \boldsymbol{0}$$

Traces = dimensions

$$\mathrm{tr}(\boldsymbol{P}) = \mathrm{tr}\left((\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{X}\right) = k$$

$$\mathrm{tr}(\boldsymbol{M}) = n - k$$

Eigenvalues are all 0 or 1; $\boldsymbol{P}$ has k ones, $\boldsymbol{M}$ has $n-k$.

Used repeatedly: $\mathrm{tr}(ABC) = \mathrm{tr}(BCA)$ (cyclic property).

Leverage ★

The ith diagonal element of $\boldsymbol{P}$,

$$h_{ii} = X_i'(\boldsymbol{X}'\boldsymbol{X})^{-1}X_i = \frac{\partial \hat{Y}_i}{\partial Y_i},$$

is the leverage of observation i: how much unit i's own outcome pulls its own fitted value.

- $0 \leq h_{ii} \leq 1$ and $\sum_i h_{ii} = k$, so average leverage is k/n.
- High leverage means unusual *covariate* values, not an unusual outcome.
- $\text{Var}(\hat{e}_i) = \sigma^2(1 - h_{ii})$: high-leverage points have artificially small residuals, they drag the line toward themselves.
- Leverage returns next week in the finite-sample corrections HC2 and HC3.

Sums of Squares, Again ★

Because $\boldsymbol{y} = \boldsymbol{P}\boldsymbol{y} + \boldsymbol{M}\boldsymbol{y}$ and $\boldsymbol{P}\boldsymbol{M} = \boldsymbol{0}$, Pythagoras gives

$$\boldsymbol{y}'\boldsymbol{y} = \boldsymbol{y}'\boldsymbol{P}\boldsymbol{y} + \boldsymbol{y}'\boldsymbol{M}\boldsymbol{y} = \hat{\boldsymbol{y}}'\hat{\boldsymbol{y}} + \hat{\boldsymbol{e}}'\hat{\boldsymbol{e}}.$$

Subtracting $n\bar{Y}^2$ (valid when a constant is included) returns the scalar decomposition $TSS = ESS + RSS$.

- R^2 is also $\cos^2$ of the angle between $\boldsymbol{y} - \bar{Y}\boldsymbol{1}$ and $\hat{\boldsymbol{y}} - \bar{Y}\boldsymbol{1}$: fit is literally about angles.
- Adding a regressor enlarges $\text{col}(\boldsymbol{X})$, which can only move $\hat{\boldsymbol{y}}$ closer to $\boldsymbol{y}$, the geometric reason R^2 never falls.

Properties

The Classical Assumptions

A1 **Linearity in parameters**: $Y_i = X_i'\beta + \epsilon_i$. (Not restrictive: X may contain logs, powers, splines, interactions.)

A2 **Random sampling**: (Y_i, X_i) i.i.d. from the population.

A3 **No perfect collinearity**: X genuinely varies; $\text{rank}(\boldsymbol{X}) = k$.

A4 **Strict exogeneity**: $E[\epsilon_i \mid X_i] = 0$.

A5 **Spherical errors**: $\text{Var}(\epsilon_i \mid \boldsymbol{X}) = \sigma^2$ for all i (homoskedasticity), and $\text{Cov}(\epsilon_i, \epsilon_j \mid \boldsymbol{X}) = 0$ for $i \neq j$ (no correlation across units).

A6 **Normality** (optional): $\epsilon_i \mid X_i \sim \mathcal{N}(0, \sigma^2)$.

What each buys you

A1–A4 $\Rightarrow$ unbiasedness. +A5 $\Rightarrow$ the simple variance formula and Gauss–Markov efficiency. +A6 $\Rightarrow$ exact t and F in finite samples.
Week 2 drops A5. Weeks 5+ interrogate A4, the only one that matters for causality.

Unbiasedness, in Scalar Form

Take the bivariate case. Start from the slope formula and substitute $Y_i = \beta_0 + \beta_1 X_i + \epsilon_i$:

$$\begin{aligned}
\hat{\beta}_1 &= \frac{\sum_i (X_i - \bar{X})(Y_i - \bar{Y})}{\sum_i (X_i - \bar{X})^2} = \frac{\sum_i (X_i - \bar{X})\big(\beta_1 (X_i - \bar{X}) + (\epsilon_i - \bar{\epsilon})\big)}{\sum_i (X_i - \bar{X})^2} \\
&= \beta_1 + \frac{\sum_i (X_i - \bar{X})\epsilon_i}{\sum_i (X_i - \bar{X})^2}.
\end{aligned}$$

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$$= \beta_1 + \frac{\sum_i (X_i - \bar{X})\epsilon_i}{\sum_i (X_i - \bar{X})^2}.$$

Remember this line

$\hat{\beta}_1 = \beta_1 +$ (a weighted average of the errors). Every result about bias and variance in this course comes from staring at it.

Taking expectations conditional on X and using A4 ($E[\epsilon_i \mid X] = 0$) gives $E[\hat{\beta}_1 \mid X] = \beta_1$, and hence $E[\hat{\beta}_1] = \beta_1$.

Unbiasedness, in Matrix Form ★

The same two lines, for any k:

$$\hat{\boldsymbol{\beta}} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'(\boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}) = \boldsymbol{\beta} + (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{\epsilon},$$

$$E[\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}] = \boldsymbol{\beta} + (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\underbrace{E[\boldsymbol{\epsilon} \mid \boldsymbol{X}]}_{=\boldsymbol{0}} = \boldsymbol{\beta}. \qquad \blacksquare$$

- Note where the work is done: *only* by A4.
- Heteroskedasticity, clustering, and non-normality do not bias $\hat{\boldsymbol{\beta}}$. They affect its *variance*, next week.
- A4 fails whenever an omitted variable, reverse causation, measurement error in X, or selection puts something correlated with X into ϵ. That is the entire causal-inference problem in one sentence.

The Variance of $\hat{\beta}_1$, in Scalar Form

From $\hat{\beta}_1 - \beta_1 = \dfrac{\sum_i (X_i - \bar{X})\epsilon_i}{\sum_i (X_i - \bar{X})^2}$, and treating X as fixed:

$$\mathrm{Var}(\hat{\beta}_1 \mid X) = \frac{\sum_i (X_i - \bar{X})^2 \ \mathrm{Var}(\epsilon_i \mid X)}{\left[\sum_i (X_i - \bar{X})^2\right]^2}$$

using only that the ϵ_i are uncorrelated across units.

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using only that the ϵ_i are uncorrelated across units.

Under homoskedasticity (A5), $\mathrm{Var}(\epsilon_i \mid X) = \sigma^2$ factors out:

$$\boxed{\mathrm{Var}(\hat{\beta}_1 \mid X) = \frac{\sigma^2}{\sum_i (X_i - \bar{X})^2}}$$

Keep the first expression in view. Next week we simply refuse to assume $\mathrm{Var}(\epsilon_i \mid X)$ is constant, and that is all a "robust standard error" is.

What Drives Precision

$$\mathrm{Var}(\hat{\beta}_1 \mid X) = \frac{\sigma^2}{\sum_i (X_i - \bar{X})^2} = \frac{\sigma^2}{n\ \mathrm{Var}_n(X)},$$

and with more regressors, $\mathrm{Var}(\hat{\beta}_j) = \dfrac{\sigma^2}{n\ \mathrm{Var}_n(X_j)\,(1 - R_j^2)}$ where R_j^2 is from regressing X_j on the other regressors.

- **Noise** σ^2 up $\Rightarrow$ variance up.
- **Sample size** n up $\Rightarrow$ variance down like $1/n$; standard error down like $1/\sqrt{n}$.
- **Spread of** X up $\Rightarrow$ variance down.
- **Collinearity** $R_j^2 \to 1 \Rightarrow$ variance explodes; the *variance inflation factor* is $1/(1 - R_j^2)$.

Collinearity violates no assumption. It says the data contain little independent information about β_j.

Estimating σ^2

$\sigma^2 = E[\epsilon_i^2]$ is unknown; we have residuals, not errors. The natural estimator is

$$s^2 = \frac{1}{n-k}\sum_i \hat{e}_i^2, \qquad E[s^2] = \sigma^2.$$

- Why $n-k$ and not n? The residuals satisfy k linear constraints (the normal equations), so they are systematically "too short."
- In matrix form ★: $\hat{\boldsymbol{e}} = \boldsymbol{M}\boldsymbol{\epsilon}$, so $E[\hat{\boldsymbol{e}}'\hat{\boldsymbol{e}}] = \sigma^2\,\mathrm{tr}(\boldsymbol{M}) = \sigma^2(n-k)$.
- Then $\hat{V}(\hat{\beta}_1) = s^2 / \sum_i (X_i - \bar{X})^2$, whose square root is the standard error your software prints.

The Gauss–Markov Theorem ★

Theorem

Under A1–A5, $\hat{\boldsymbol{\beta}}$ is the Best Linear Unbiased Estimator: among all $\tilde{\boldsymbol{\beta}} = \boldsymbol{C}\boldsymbol{y}$ with $\boldsymbol{C}$ a function of $\boldsymbol{X}$ and $E[\tilde{\boldsymbol{\beta}} \mid \boldsymbol{X}] = \boldsymbol{\beta}$, $\mathrm{Var}(\tilde{\boldsymbol{\beta}} \mid \boldsymbol{X}) - \mathrm{Var}(\hat{\boldsymbol{\beta}} \mid \boldsymbol{X}) \succeq 0$.

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Proof. Write $\boldsymbol{C} = (\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' + \boldsymbol{D}$. Unbiasedness for all β forces $\boldsymbol{D}\boldsymbol{X} = \boldsymbol{0}$. Then

$$\begin{aligned}\mathrm{Var}(\boldsymbol{C}\boldsymbol{y} \mid \boldsymbol{X}) &= \sigma^2\left[(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}' + \boldsymbol{D}\right]\left[\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1} + \boldsymbol{D}'\right] \\ &= \sigma^2(\boldsymbol{X}'\boldsymbol{X})^{-1} + \sigma^2\underbrace{(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{D}'}_{=\boldsymbol{0}} + \sigma^2\underbrace{\boldsymbol{D}\boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}}_{=\boldsymbol{0}} + \sigma^2\boldsymbol{D}\boldsymbol{D}' \\ &= \mathrm{Var}(\hat{\beta} \mid \boldsymbol{X}) + \sigma^2\boldsymbol{D}\boldsymbol{D}' \succeq \mathrm{Var}(\hat{\beta} \mid \boldsymbol{X}). \qquad \blacksquare\end{aligned}$$

Reading Gauss–Markov Correctly

- "Best" only within the class of linear unbiased estimators. Biased estimators (ridge, lasso, shrinkage) can have far lower mean squared error, Week 12.
- Efficiency requires A5. Under heteroskedasticity OLS stays unbiased but is no longer BLUE; weighted least squares with the correct weights is.
- In practice we rarely know the weights, so we keep OLS and fix the *standard errors* instead, the entire subject of next week.
- Gauss–Markov says nothing about causality. It is a statement about the variance of an estimator for the population regression coefficient, whatever that coefficient happens to mean.

Finite-Sample Distribution Under Normality

If A6 also holds, then conditional on X,

$$\hat{\beta}_1 \sim \mathcal{N}\Big(\beta_1,\ \tfrac{\sigma^2}{\sum_i (X_i - \bar{X})^2}\Big), \qquad \frac{(n-k)s^2}{\sigma^2} \sim \chi^2_{n-k}, \qquad \hat{\beta}_1 \perp\!\!\!\perp s^2,$$

so that $t = \dfrac{\hat{\beta}_1 - \beta_1}{\mathrm{se}(\hat{\beta}_1)} \sim t_{n-k}$.

- $\hat{\beta}_1$ is a *linear* function of the errors, hence normal when they are.
- Without A6, all of this holds only *approximately*, and only for large n. That is next week.

Partialling Out

The Frisch–Waugh–Lovell Theorem ★

Partition $\boldsymbol{X} = [\,\boldsymbol{X}_1 \quad \boldsymbol{X}_2\,]$: a treatment of interest and controls. Let $\boldsymbol{M}_2 = \boldsymbol{I} - \boldsymbol{X}_2(\boldsymbol{X}_2'\boldsymbol{X}_2)^{-1}\boldsymbol{X}_2'$ and define

$$\tilde{\boldsymbol{X}}_1 = \boldsymbol{M}_2\boldsymbol{X}_1, \qquad \tilde{\boldsymbol{y}} = \boldsymbol{M}_2\boldsymbol{y}.$$

Theorem

The OLS estimate $\hat{\beta}_1$ from the full regression equals

$$\hat{\beta}_1 = (\tilde{\boldsymbol{X}}_1'\tilde{\boldsymbol{X}}_1)^{-1}\tilde{\boldsymbol{X}}_1'\tilde{\boldsymbol{y}} = (\tilde{\boldsymbol{X}}_1'\tilde{\boldsymbol{X}}_1)^{-1}\tilde{\boldsymbol{X}}_1'\boldsymbol{y},$$

and the residuals from the short regression are numerically identical to $\hat{\boldsymbol{e}}$ from the full one.

This is the three-step recipe from earlier, now for any number of controls.

Proof of FWL ★

Premultiply the model by $\boldsymbol{M}_2$ and use $\boldsymbol{M}_2\boldsymbol{X}_2 = \boldsymbol{0}$:

$$\boldsymbol{M}_2\boldsymbol{y} = \boldsymbol{M}_2\boldsymbol{X}_1\boldsymbol{\beta}_1 + \underbrace{\boldsymbol{M}_2\boldsymbol{X}_2}_{=\boldsymbol{0}}\boldsymbol{\beta}_2 + \boldsymbol{M}_2\boldsymbol{\epsilon} \quad \Longrightarrow \quad \tilde{\boldsymbol{y}} = \tilde{\boldsymbol{X}}_1\boldsymbol{\beta}_1 + \boldsymbol{M}_2\boldsymbol{\epsilon}.$$

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For the estimator, decompose the fitted model $\boldsymbol{y} = \boldsymbol{X}_1\hat{\boldsymbol{\beta}}_1 + \boldsymbol{X}_2\hat{\boldsymbol{\beta}}_2 + \hat{\boldsymbol{e}}$ and premultiply by $\tilde{\boldsymbol{X}}_1' = \boldsymbol{X}_1'\boldsymbol{M}_2$:

$$\tilde{\boldsymbol{X}}_1'\boldsymbol{y} = \tilde{\boldsymbol{X}}_1'\tilde{\boldsymbol{X}}_1\hat{\boldsymbol{\beta}}_1 + \boldsymbol{X}_1'\underbrace{\boldsymbol{M}_2\boldsymbol{X}_2}_{=\mathbf{0}}\hat{\boldsymbol{\beta}}_2 + \underbrace{\boldsymbol{X}_1'\boldsymbol{M}_2\hat{\boldsymbol{e}}}_{=0},$$

using $\boldsymbol{M}_2\hat{\boldsymbol{e}} = \hat{\boldsymbol{e}}$ (since $\boldsymbol{X}_2'\hat{\boldsymbol{e}} = \mathbf{0}$) and $\boldsymbol{X}_1'\hat{\boldsymbol{e}} = \mathbf{0}$. Solving gives the result. Using $\tilde{\boldsymbol{y}}$ instead of $\boldsymbol{y}$ changes nothing because $\tilde{\boldsymbol{X}}_1'\boldsymbol{M}_2 = \tilde{\boldsymbol{X}}_1'$. ■

Why FWL Matters More Than It Looks

1. **Interpretation.** "Controlling for W" = discarding all covariance between X_1 and W. Your estimate lives on what remains.
2. **Computation.** Fixed-effects estimators *are* FWL: regressing on unit dummies is the same as demeaning within unit (Week 10).
3. **Diagnostics.** The added-variable plot is the only two-dimensional picture that faithfully represents a multivariate slope.
4. **Generality.** Nothing in the argument requires $\boldsymbol{X}_2$ to enter linearly, it only requires that we can compute the part of X_1 that $\boldsymbol{X}_2$ predicts. That observation is the seed of a large literature, and we return to it briefly in Week 13.

Omitted Variables, in Matrix Form ★

With $\boldsymbol{X}_1$ included and $\boldsymbol{X}_2$ omitted,

$$E[\tilde{\boldsymbol{\beta}}_1 \mid \boldsymbol{X}] = \boldsymbol{\beta}_1 + \underbrace{(\boldsymbol{X}_1'\boldsymbol{X}_1)^{-1}\boldsymbol{X}_1'\boldsymbol{X}_2}_{\hat{\boldsymbol{\delta}}}\,\boldsymbol{\beta}_2.$$

- Each column of $\hat{\boldsymbol{\delta}}$ is the coefficient vector from regressing one omitted variable on all the included ones, exactly δ_1 from the scalar case, stacked.
- With several omitted variables the bias is a *sum* of terms which may offset, so "the bias must be positive" arguments require care.
- Sensitivity analysis (Week 6) formalizes the question: how strong would an omitted confounder have to be to overturn the result?

Adjustment

Regression With a Binary Treatment

Let $D_i \in \{0, 1\}$ be a treatment and W_i controls:

$$Y_i = \alpha + \rho D_i + W_i'\gamma + e_i.$$

Suppose that *within cells of* W_i treatment is as good as randomly assigned. Then the cell-specific difference in means

$$\tau(w) = E[Y_i \mid D_i = 1, W_i = w] - E[Y_i \mid D_i = 0, W_i = w]$$

is the causal effect in that cell.

Question: when $\tau(w)$ varies across cells, what single number does OLS report?

OLS Reports a Variance-Weighted Average ★

If the model is saturated in W_i (Angrist and Pischke 2009):

$$\rho = \frac{E\big[\sigma_D^2(W_i)\,\tau(W_i)\big]}{E\big[\sigma_D^2(W_i)\big]}, \qquad \sigma_D^2(w) = \mathrm{Var}(D_i \mid W_i = w) = p(w)\big(1 - p(w)\big),$$

where $p(w) = \Pr(D_i = 1 \mid W_i = w)$ is the *propensity score.*

- OLS averages cell-level effects with weights proportional to the conditional variance of treatment.
- Cells with $p(w) \approx 0.5$ dominate; cells where nearly everyone is (or is not) treated get almost no weight.
- So OLS does not generally deliver the ATE or the ATT, it delivers its own weighted average. Matching and weighting (Week 6) let you choose the weights deliberately.

Good Controls, Bad Controls

Adding a control is not automatically an improvement.

Good controls

- Confounders: common causes of D and Y.
- Pre-treatment covariates that predict Y (precision gains).

Bad controls

- Post-treatment variables (mediators): conditioning removes part of the effect you want.
- Colliders: common effects of D and Y; conditioning *creates* association where none existed.
- Proxies for the outcome.

Preview

No algebra distinguishes these cases: the distinction is *causal*, not statistical. We need a language for causal assumptions, which is Week 5.

Where This Leaves Us

- OLS minimizes squared errors. Everything else, the slope formula, the normal equations, FWL, OVB, follows from that one criterion by algebra.
- In the population, OLS estimates the best linear approximation to the CEF. Under A1–A4 the sample version is unbiased for it.
- Whether that coefficient is a *causal effect* is a separate question that regression cannot answer.
- **Next week**: we have $\hat{\beta}$; now we need to know how uncertain it is, and the standard error your software prints is usually the wrong one.

Before Thursday

Install `R` and `RStudio`; skim MHE Chapter 3 and CCI Chapter 6. Lab covers the matrix algebra used today and OLS by hand.

References

- Angrist, Joshua D. and Jörn-Steffen Pischke. 2009. *Mostly Harmless Econometrics*. Princeton University Press. [Ch. 3]
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- Correll, Josh, Abigail M. Folberg, Charles M. Judd, Gary H. McClelland, and Carey S. Ryan. 2024. *Data Analysis: A Model Comparison Approach to Regression, ANOVA, and Beyond*, 4th ed. Routledge. [for review]